

Zombies can gain by waiting: a 59-vertex cactus whose zombie number drops when a leaf is attached (a negative answer to Question 4.1 of Bartier, Bénéteau, Bonamy, La and Narboni)

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Machine-generated note

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Abstract

In the deterministic game of zombies and survivor of Fitzpatrick, Howell, Messinger and Pike, Bartier, Bénéteau, Bonamy, La and Narboni asked (Question 4.1 of arXiv:2008.03587) whether attaching vertices of degree one to a graph can ever change its zombie number z ; equivalently, whether zombies can gain by waiting before they start to move. We answer this in the negative: an explicit cactus G on 59 vertices, consisting of three odd cycles of lengths 25, 17 and 9 hung from a path of length 10, has $z(G) \geq 3$, while the graph H obtained by attaching one leaf to the far end of the path has $z(H) = 2$. The proof is elementary and self-contained: a rotation argument on a pendant odd cycle (in substance the survivor’s “Busan strategy” of the source paper) shows that two zombies never suffice on G , and a bracketing argument on the same cycles shows that two zombies placed at the two ends of the extended path capture the survivor on H , the leaf delaying the rear zombie by exactly the one move that the three cycle lengths $25 = 2 \cdot 11 + 3$, $17 = 2 \cdot 7 + 3$, $9 = 2 \cdot 3 + 3$ are tuned to. An exact game solver run by the referee gives $z(G) = 3$ and $z(H) = 2$, the placement used here being the unique winning pair on H .

This note was produced by AI within the [Graph Theory AI](#) project: the argument was found by a language model and audited by an adversarial LLM referee, and it has not been checked by a human mathematician. [Section 5](#) records the full provenance.

1 Introduction

The game. Zombies and Survivor is the variant of Cops and Robber in which the pursuers are compelled to approach. We use the deterministic rules of Fitzpatrick, Howell, Messinger and Pike [2], exactly as restated in Section 1 of the source paper [1]:

- (R1) all graphs are finite and connected, and there is a single survivor;
- (R2) the zombies are placed first, then the survivor chooses a starting vertex;
- (R3) in each round every zombie moves first, along an edge that lies on a shortest path (geodesic) from its vertex to the survivor’s; this move is *compulsory*, and when several geodesics are available the zombie *chooses* which edge to take (this free will is what “deterministic” refers to);
- (R4) the survivor then moves to a neighbouring vertex or stays where it is;
- (R5) the zombies capture the survivor if a zombie moves onto the survivor’s vertex.

The *zombie number* $z(G)$ is the least k such that k zombies have an initial placement from which they can force capture against every starting vertex and every strategy of the survivor. Since a zombie strategy is in particular a cop strategy, $c(G) \leq z(G)$ where c is the cop number; the source paper exhibits graphs with $c = 2$ and z arbitrarily large. As in the referee’s solver we regard a survivor who

moves onto a vertex occupied by a zombie as captured (the zombie then stands at distance zero); the survivor strategies below never make such a move, so the convention only affects the bookkeeping in the upper bound (see [Remark 4.2](#)).

The problem. Because a zombie starting on a pendant path must spend its first moves walking to the end of that path, attaching leaves to a graph lets some zombies “wait” before entering the original graph. The source paper asks whether this can ever matter.

Question 1.1 ([1, Question 4.1]). For every graph G , and for a graph G' obtained from G by successively adding vertices of degree 1, does it always hold that $z(G') = z(G)$?

The authors add that the question “can be interpreted as: is there any advantage for zombies to individually wait for some pre-announced time at the beginning of the game (and then activate and follow the standard rules)?” [1]. We answer the question in the negative, by an explicit example in which waiting helps.

Theorem 1.2 (informal; [Theorem 2.2](#)). *There is a cactus G on 59 vertices with $z(G) \geq 3$ such that the graph H obtained from G by attaching a single vertex of degree one has $z(H) = 2$. In fact $z(G) = 3$.*

Related work. Deterministic zombies were introduced in [2]. The source paper [1] proves $z(G \square H) \leq z(G) + z(H)$ for the Cartesian product and gives a family with cop number 2 and unbounded zombie number; its Section 3 contains the survivor’s “Busan strategy”, a rotation around a long cycle, which is in substance the rotation lemma below ([Lemma 3.3](#)). According to the referee’s search, only two works cite [1]: Bose, De Carufel and Shermer [3], on zombie and lazy-zombie numbers of graph classes, and Liu and Miller [4], who determine the zombie number of “foliage graphs” (cycles sharing one vertex); Lemma 1 of [4] is close to our [Lemma 3.3](#). Neither addresses pendant vertices or [Question 1.1](#), and no resolution of the question was found elsewhere ([Remark 4.8](#)).

An earlier machine attempt on this problem (by gpt-5.6-sol; supplied to the model as context, see [Section 5](#)) proved the general inequality $z(G') \leq z(G)$ for every pendant extension G' of G , and showed that a counterexample to [Question 1.1](#) must have $z(G) \geq 3$ and $c(G) < z(G)$ and must use a mixture of zombies that start inside G and zombies that start on the attached leaves. We do not reprove these facts, and none of them is needed below; the example of this note falls exactly in the window they leave open ([Remark 4.5](#)).

Notation. $d_Q(u, w)$ is the graph distance in Q . For $n \geq 1$ and $a \in \mathbb{Z}$ or $a \in \mathbb{Z}_n$ we write $\|a\|_n = \min\{|a'| : a' \in \mathbb{Z}, a' \equiv a \pmod{n}\}$ for the cyclic distance from a to 0 modulo n ; thus $\|a\|_n \leq \lfloor n/2 \rfloor$, and if $n = 2h + 1$ then $\|a\|_n \leq h$. A *round* consists of the zombies’ moves followed by the survivor’s move; “the start of round $t + 1$ ” is the moment after t complete rounds, before the zombies move.

2 The two graphs and the main theorem

Definition 2.1 (the graphs G and H). Let P be the path $v_0 v_1 \cdots v_{10}$. The graph G is obtained from P by attaching three cycles through three of its vertices, with all other cycle vertices new and pairwise distinct:

cycle	length	vertices	edges
A	25	$a_0 = v_0, a_1, \dots, a_{24}$	$a_i a_{i+1}$ for $i \in \mathbb{Z}_{25}$
B	17	$b_0 = v_2, b_1, \dots, b_{16}$	$b_i b_{i+1}$ for $i \in \mathbb{Z}_{17}$
C	9	$c_0 = v_4, c_1, \dots, c_8$	$c_i c_{i+1}$ for $i \in \mathbb{Z}_9$

So $V(G) = \{v_0, \dots, v_{10}\} \cup \{a_1, \dots, a_{24}\} \cup \{b_1, \dots, b_{16}\} \cup \{c_1, \dots, c_8\}$, $|V(G)| = 11 + 24 + 16 + 8 = 59$, and $E(G)$ consists of the ten path edges $v_i v_{i+1}$ ($0 \leq i \leq 9$) together with the $25 + 17 + 9$ cycle edges of the table; $|E(G)| = 61$. There are no other edges. G is connected and is a cactus (every edge lies on at most one cycle; $|E| = |V| - 1 + 3$).

The graph H is obtained from G by adding one new vertex v_{11} and the single edge $v_{10} v_{11}$; so v_{11} has degree one, $|V(H)| = 60$, $|E(H)| = 62$, and H is obtained from G by adding a vertex of degree 1 in the sense of [Question 1.1](#).

The three cycles have odd lengths $25 = 2 \cdot 12 + 1$, $17 = 2 \cdot 8 + 1$, $9 = 2 \cdot 4 + 1$, which is what the lower bound uses; they also satisfy $25 = 2 \cdot 11 + 3$, $17 = 2 \cdot 7 + 3$, $9 = 2 \cdot 3 + 3$ with $11 = 11 - 2 \cdot 0$, $7 = 11 - 2 \cdot 2$, $3 = 11 - 2 \cdot 4$ for the attachment indices $0, 2, 4$, which is what the upper bound uses ([Remark 4.3](#)).

Theorem 2.2 (main theorem). *Let G and H be the graphs of [Definition 2.1](#). Then $z(G) \geq 3$ and $z(H) = 2$. More precisely:*

- (i) *for every choice of two starting vertices in G (not necessarily distinct) the survivor has a starting vertex and a strategy that avoids capture forever;*
- (ii) *in H , two zombies placed at v_0 and v_{11} capture the survivor from every starting vertex, within at most 17 rounds;*
- (iii) *in H , one zombie can be evaded from every starting vertex.*

Corollary 2.3. *The answer to [Question 1.1](#) is negative: H is obtained from G by adding one vertex of degree 1, yet $z(H) = 2 < 3 \leq z(G)$. In particular zombies can gain by waiting: the winning placement on H consists of one zombie that is active at once and one zombie that is delayed by exactly one move.*

Proof. Immediate from [Theorem 2.2](#): (i) gives $z(G) \geq 3$, (ii) gives $z(H) \leq 2$ and (iii) gives $z(H) \geq 2$. The zombie at v_{11} must spend its first move going to v_{10} whatever the survivor does, so its play on H is that of a zombie placed at v_{10} in G that waits one round. $\square$

Idea of proof. Both halves rest on two facts about a cycle Z of odd length hung from a cut vertex ([Section 3](#)).

Lower bound. If Z has length $2h + 1$ and a zombie is on Z at forward distance $g \in \{2, \dots, h\}$ behind a survivor who keeps walking around Z , the zombie's only legal move is to follow, so g never changes and the survivor is never caught; a zombie outside Z is forced to walk to the root of Z and, on arrival, finds itself at a forward distance determined by its initial distance to the root (its *phase*). Two zombies can both be kept in the window $\{2, \dots, h\}$ as soon as their phases differ by at most $h - 2$ cyclically ([Lemma 3.3](#) and [Corollary 3.4](#)): 10 on A , 6 on B , 2 on C . A short case analysis on where the two zombies start (both outside A , both on A , or one of each) shows that one of the three cycles always accommodates both phases, because every vertex outside A is within distance 10 of v_0 , the vertices of A are within 12, and the only bad pair on A (distances 12 and 1 from v_0 , the latter forcing the zombie to be at v_1) is rescued by B . Hence two zombies never win on G ; a single zombie is evaded on A in the same way, in G or in H .

Upper bound. Place the zombies at v_0 and v_{11} in H and let them walk along the path towards the survivor's projection on the path. Since 11 is odd they never collide; after t rounds they stand at v_t and v_{11-t} with the survivor's projection strictly between them, which is impossible for $t = 5$ (v_5, v_6 are adjacent), so by round 5 either the survivor is caught on the path or the front zombie reaches the root v_r ($r \in \{0, 2, 4\}$) of the cycle containing the survivor, at time $t = r$, with the rear zombie at v_{11-r} , at distance $d = 11 - 2r$ from the root; the rear zombie, which stands at v_{11-t} with $11 - t \geq 6$, never makes such a contact first. The cycle then has length exactly $2d + 3$, and the capture lemma ([Lemma 3.5](#)) finishes: the root zombie chases the survivor around the cycle, which keeps the survivor away from the root for the d rounds the second zombie needs to arrive at the root; then the survivor is bracketed between two zombies approaching from opposite sides along an arc, and the arc loses two edges per round. [\[full proof\]](#)

3 Two lemmas about a cycle hanging from a cut vertex

Throughout this section Q is a finite connected graph.

Definition 3.1 (pendant cycle). A *pendant cycle* of Q with root ρ is an induced cycle Z of Q together with a vertex $\rho \in V(Z)$ such that ρ is the only vertex of Z having a neighbour outside $V(Z)$. When Z has length n we fix an orientation and label $V(Z)$ by $\mathbb{Z}_n$ along it, with $\rho = 0$; the label of a vertex is its *coordinate*. For a vertex x of Q we define its *phase* on Z by

$$p(x) = \begin{cases} x & \text{if } x \in V(Z) \text{ (its coordinate),} \\ -d_Q(x, 0) \bmod n & \text{if } x \notin V(Z). \end{cases}$$

In G and in H the cycles A, B, C are pendant cycles with roots v_0, v_2, v_4 respectively: each is induced, and its root is its only vertex with neighbours outside it. The following lemma records the metric facts that the writeup used tacitly (the referee's Caveat C1, [Appendix B](#)).

Lemma 3.2 (distances around a pendant cycle). *Let Z be a pendant cycle of Q of length n , labelled by $\mathbb{Z}_n$ with root 0.*

- (a) Z is isometric in Q : $d_Q(u, w) = \|u - w\|_n$ for all $u, w \in V(Z)$.
- (b) For $y \notin V(Z)$ and $s \in V(Z)$: $d_Q(y, s) = d_Q(y, 0) + \|s\|_n$.
- (c) For $y \notin V(Z)$ and $s \in V(Z)$, a neighbour y' of y satisfies $d_Q(y', s) = d_Q(y, s) - 1$ if and only if $d_Q(y', 0) = d_Q(y, 0) - 1$. Consequently, while the survivor is on Z , every legal move of a zombie outside Z decreases its distance to the root by exactly one, whichever geodesic it chooses, and the zombie reaches the root after exactly $d_Q(y, 0)$ moves.
- (d) Suppose $n = 2h + 1$ is odd. If $c, s \in V(Z)$ and $s - c \equiv g \pmod{n}$ with $1 \leq g \leq h$, then $d_Q(c, s) = g$ and $c + 1$ is the unique neighbour of c (in Q) closer to s than c ; symmetrically, if $s - c \equiv -g \pmod{n}$ with $1 \leq g \leq h$ then $c - 1$ is the unique closer neighbour.

Idea of proof. Every path from $V(Z) \setminus \{0\}$ to the outside of Z passes through the root, so a shortest path between two cycle vertices never leaves Z , and distances from outside to the cycle split at the root. Part (d) is the arithmetic of $\min(g, n - g)$ for odd n . [\[full proof\]](#)

Lemma 3.3 (rotation lemma). *Let Z be a pendant cycle of Q of odd length $n = 2h + 1$ with $h \geq 2$, labelled by $\mathbb{Z}_n$ with root 0. Suppose k zombies start at vertices $x_1, \dots, x_k$ of Q and that some $q \in \mathbb{Z}_n$ satisfies*

$$q - p(x_i) \bmod n \in \{2, 3, \dots, h\} \quad \text{for every } i = 1, \dots, k. \quad (1)$$

Then the survivor who starts at the vertex q of Z and, in every round, moves to the next vertex in the orientation is never captured.

Corollary 3.4. *In the setting of [Lemma 3.3](#):*

- (i) *one zombie, wherever it starts, can be evaded forever (take $q = p(x_1) + 2$);*
- (ii) *two zombies starting at x_1, x_2 can be evaded forever whenever $\|p(x_1) - p(x_2)\|_n \leq h - 2$.*

For the three cycles of G : $h - 2 = 10$ on A ($n = 25, h = 12$), $h - 2 = 6$ on B ($n = 17, h = 8$), $h - 2 = 2$ on C ($n = 9, h = 4$).

Idea of proof. A zombie on Z at forward gap $g \in \{2, \dots, h\}$ behind the survivor has, by [Lemma 3.2\(d\)](#), exactly one legal move, one step forward; the survivor also steps forward, so g is invariant and there is never contact. A zombie outside Z at distance δ from the root loses one unit of δ per round ([Lemma 3.2\(c\)](#)) while the survivor's coordinate gains one, so the sum “coordinate of survivor + δ ” is invariant; (1) says this sum is in $\{2, \dots, h\}$, so when δ reaches 0 the survivor is not at the root and the zombie enters the first regime with a gap in the window. For the corollary, the differences of two elements of $\{2, \dots, h\}$ fill exactly $[-(h - 2), h - 2]$. [\[full proof\]](#)

The rotation strategy is, in substance, the survivor’s “Busan strategy” of [1, Section 3], and is close to Lemma 1 of Liu and Miller [4]; we reprove it because the precise phase condition (1) for zombies starting *outside* the cycle is what the case analysis on G needs.

Lemma 3.5 (capture lemma). *Let Z be a pendant cycle of Q of length $n = 2d + 3$ with $d \geq 1$ and root 0. Suppose that at the start of some round one zombie stands at the root, a second zombie stands at a vertex $y \notin V(Z)$ with $d_Q(y, 0) = d$, and the survivor stands at a vertex of $V(Z) \setminus \{0\}$. Then the two zombies can force capture (within $d + \lceil d/2 \rceil + 1$ rounds).*

Idea of proof. Orient Z so that the survivor’s coordinate p lies in $[1, d + 1]$ (possible as n is odd). The root zombie steps forward every round; its forward gap to the survivor, which starts at $p \leq d + 1$, can only decrease, so by Lemma 3.2(d) the forward step is always its unique legal move, and while the gap is at least 2 the survivor’s unwrapped coordinate stays in $[t + 2, t + d + 1] \subset (0, n)$ after $t \leq d$ rounds: the survivor cannot reach the root and cannot leave the cycle. Meanwhile the outside zombie is forced to the root (Lemma 3.2(c)) and arrives after exactly d rounds. Now the zombies stand at coordinates d and $n \equiv 0$ with the survivor strictly between them on the arc from d to n , both arc distances at most $d + 1$ and therefore genuine shortest-path distances; both zombies step inward, the arc between them loses two edges per round, the survivor cannot pass through an occupied endpoint, and capture follows. [full proof]

4 Remarks

Remark 4.1 (the referee’s exact computation). The referee (Appendix B) wrote an exact solver for the game (attractor computation over states “ordered zombie tuple, survivor, zombies to move”, existential over the zombies’ joint choice of geodesics and universal over the survivor’s replies), validated it on graphs with known zombie number (K_2 , K_3 , cycles $C_4, \dots, C_{11}$, paths, $K_{1,5}$, the Petersen graph) and cross-checked it against an independently written fixed-point solver from an earlier campaign on 120 random connected graphs on 4–9 vertices with 0 mismatches. On the graphs of Definition 2.1 it found: no 1- or 2-zombie placement wins on G , and exactly 1250 unordered triples win, so $z(G) = 3$ exactly (the writeup and Theorem 2.2 only claim $z(G) \geq 3$); no 1-zombie placement wins on H , and exactly one unordered pair wins on H , namely $\{v_0, v_{11}\}$, which beats all 60 survivor starts, so $z(H) = 2$ and the placement of Theorem 2.2(ii) is the *only* one that works. The referee also enumerated all $\binom{59}{2} + 59 = 1770$ unordered pairs of starting vertices in G and confirmed that each admits a rotation certificate (Corollary 3.4(ii)) on A , B or C , i.e. that the three-case analysis in the proof of Theorem 2.2(i) is exhaustive with no uncovered pair, and it verified the distance facts used there ($\max d(v_0, \cdot) = 10$ outside A and 12 on A ; v_1 the only vertex outside A at distance 1 from v_0 ; $d(x, v_2) = d(x, v_0) + 2$ and $d(x, v_4) = d(x, v_0) + 4$ for $x \in A$). Scripts: `verification_astra/scripts/2008.03587__00/`.

Remark 4.2 (rule conventions). The proofs use the rules (R1)–(R5) of [2] as restated in [1]; the referee checked this restatement against the TeX source of [1] (lines 165–190) and confirmed in particular that the zombie team exercises free will over which geodesic to follow, in its own favour (the proof of Theorem 2.1 there has zombies “choosing to keep the same coordinate”), while survivor strategies must work against *all* zombie choices, as ours do. Two details are conventions rather than rules. First, a survivor who moves onto a zombie’s vertex is regarded as captured; the survivor strategies of Theorem 2.2(i),(iii) never do this, so the lower bounds do not depend on it, and in the upper bound it only serves to end the play at once rather than one zombie move later. Second, the referee re-ran its solver under three variants: the standard rules; “the survivor may not stay” (must move); and “the survivor moves first, then the zombies”. Under all three, two zombies do not win on G , one zombie does not win on H , and $\{v_0, v_{11}\}$ wins on H . The separation is thus insensitive to these conventions. What is essential is the compulsory motion of the zombies: under the “zombies may pass” variant z_{pass} considered by the earlier attempt, the delay mechanism disappears (that attempt proved $z_{\text{pass}}(G) \leq z(G') \leq z(G)$ for every pendant extension G').

Remark 4.3 (tightness of the parameters). The construction has no slack. The upper bound needs the cycle rooted at v_r to have length exactly $2(11 - 2r) + 3$, which gives 25, 17, 9 for $r = 0, 2, 4$; the lower bound needs the windows $h - 2 = 10, 6, 2$ of these three cycles to cover all phase differences that arise. The referee tested the same design with a path $v_0 \cdots v_L$, leaf v_{L+1} and cycles of lengths $2(L + 1 - 2r) + 3$ at $r \in \{0, 2, 4\}$: for $L = 6$ (cycles 17, 9; 31 vertices) and $L = 8$ (cycles 21, 13, 5; 45 vertices) two zombies already win on G , so there is no separation; $L = 10$ is the first case that works. Deleting any one of the three cycles from the 59-vertex G (any of the six proper subsets of $\{v_0, v_2, v_4\}$) also lets two zombies win on G . Every ingredient is load-bearing.

Remark 4.4 (what the proof uses about the host graph). The writeup stated the rotation and capture lemmas for a cycle “attached to the rest of the graph only at its vertex 0”. What the arguments actually use is [Lemma 3.2](#): the cycle is isometric and distances from outside split additively at the root. Both follow from the root being a cut vertex separating the rest of the cycle from the rest of the graph and from the cycle being induced, which is [Definition 3.1](#); in G and H this holds by construction. In the upper bound the same additivity is used along the path: for a path vertex v_i and any vertex s of H , $d_H(v_i, s) = |i - \iota(s)| + d_H(v_{\iota(s)}, s)$, where $v_{\iota(s)}$ is the projection of s to the path ([Appendix A](#)).

Remark 4.5 (relation to the earlier attempt). The earlier attempt ([Section 5](#)) established, for every pendant extension G' of any connected G , that $z(G') \leq z(G)$ and $c(G') = c(G)$, and that a strict inequality $z(G') < z(G)$ requires $z(G) \geq 3$, $c(G) < z(G)$, and a winning placement mixing zombies that start in G with zombies that start on the attached vertices. The present example is consistent with all of this: $z(G) = 3$, $z(H) = 2$, the referee notes $c(G) = 2$ for this cactus, and the unique winning placement $\{v_0, v_{11}\}$ mixes one active and one delayed zombie. The earlier attempt also showed that equal delays for all zombies never help, so a mixed placement is not merely sufficient but necessary; here the delay is the smallest possible, one move.

Remark 4.6 (what remains open). [Theorem 2.2](#) refutes the equality $z(G') = z(G)$ but drops the zombie number by exactly one. The natural refined questions are open: how far below $z(G)$ can $z(G')$ fall, and is the drop bounded by a function of $z(G)$ or unbounded? Nothing here bears on [Question 4.2](#) of [\[1\]](#). On the positive side, the earlier attempt’s inequality $z(G') \leq z(G)$ shows that leaves can only help the zombies, never the survivor.

Remark 4.7 (the sibling target and the faithfulness of the statement). The same source paper’s [Question 4.2](#) was the target 2008.03587__01 of the first campaign; that attempt was rejected by its referee because the catalog transcription had replaced a “ $\geq$ ” by “ $>$ ” and the model refuted the garbled statement. For this reason the referee of the present target began by re-downloading the arXiv TeX source of [\[1\]](#) (v3, main.tex, lines 420–423) and found that [Question 4.1](#) is transcribed verbatim in the problem statement given to the model; [Question 1.1](#) above reproduces it. The writeup attacks the genuine question, and the paper’s own gloss (waiting zombies) is exactly the mechanism of [Corollary 2.3](#).

Remark 4.8 (novelty and attribution). The referee found, via Semantic Scholar, exactly two works citing [\[1\]](#): Bose, De Carufel and Shermer [\[3\]](#) (whose arXiv source contains no occurrence of “pendant”, “degree 1” or “degree one”) and Liu and Miller [\[4\]](#) (read in full; it characterises the zombie number of foliage graphs and asks for the characterisation at $z > 2$, without mentioning pendant vertices or [Question 1.1](#)). Targeted web searches found no resolution of the question (one search-engine summary asserting an affirmative resolution was unsupported by any document and was discarded). To the referee’s knowledge the negative answer is new. The ingredients are not: the rotation strategy is the Busan strategy of [\[1, Section 3\]](#) and is close to [\[4, Lemma 1\]](#), and the writeup claimed no novelty for it. What is new is the combination: three odd cycles whose lengths are tuned to the arrival times of a one-move-delayed zombie so that a two-zombie capture exists on H , while the same lengths leave a rotation window on some cycle for every pair of starts on G .

5 Provenance

The mathematics in this note was produced without human intervention, in three stages. (1) The construction and proof were found by the language model gpt-6-astra (reasoning effort max) in a single autonomous pass started on 2026-09-17, as part of the retry leg of a sweep over open problems catalogued from arXiv (catalog id 2008.03587__00, leg attacks_retry; original writeup in attacks_retry/2008.03587__00/output.md). This was a *second* attempt on the problem: the prompt contained, as context, the earlier attempt attacks/2008.03587__00 by gpt-5.6-sol, whose verdict was “partial” (it proved $z(G') \leq z(G)$, the invariance for $z(G) \leq 2$ and for $z(G) = c(G)$, and the structural constraints recalled in Remark 4.5, but found no counterexample). The new writeup shares with it only the statement of the rules; its graphs, lemmas and proofs are constructed from scratch, and the referee confirmed that nothing was inherited uncritically. The model itself claimed the verdict “disproved” with high confidence, flagged that it relied on the compulsory-move rules with zombies moving first, and made no computational or literature claim. (2) An independent adversarial referee agent (claude-opus-5) reviewed the writeup on 2026-09-17. It re-downloaded the arXiv TeX source of the source paper and checked the statement of Question 4.1 and the game rules word for word; re-derived each of the nineteen steps it extracted from the writeup and found all of them valid, two of them (the Section 5 sentences repaired below) under-argued; wrote an exact game solver from scratch, validated it on known values and cross-checked it against an independent solver on 120 random graphs; computed $z(G) = 3$ and $z(H) = 2$ exactly with the counts in Remark 4.1; certified all 1770 start pairs on G ; tested three rule conventions and the tightness of the parameters (Remarks 4.2 and 4.3); and searched the literature (Remark 4.8). Its verdict was CONFIRMED with high confidence. Its report is reproduced verbatim in Appendix B. (3) On 2026-09-17 the writeup was rewritten into the present note by claude-fable-5-1. The deviations from the writeup are the following; no mathematical content was changed.

- The text was reorganised: the graphs are a definition with explicit vertex names and edge rules (the writeup described them in prose), the result is Theorem 2.2 with the three parts (i)–(iii) separated, Corollary 2.3 states the answer to the question, and full proofs are collected in Appendix A. The edge counts 61 and 62 and the cactus property are from the referee report.
- Definition 3.1 (pendant cycle, phase) and Lemma 3.2 were added to state explicitly the isometry of each cycle and the additivity of distances at its root, which the writeup’s rotation lemma, capture lemma and Section 5 used tacitly (the referee’s Caveat C1). The phase is now defined by the writeup’s formula but with a named orientation convention.
- The proof of Lemma 3.3 was rewritten as an explicit round-by-round invariant (survivor coordinate plus distance of an outside zombie to the root is constant), and the proof of Corollary 3.4 gives the explicit choice of g ; the writeup’s phrase “unique shortest route is in the survivor’s direction” is now Lemma 3.2(d).
- In the proof of Lemma 3.5 the writeup’s invariant $2 \leq g_t$ was relaxed to $1 \leq g_t$, with $g_t = 1$ handled as capture in the next zombie move; this removes a tacit “no capture in the following round” hypothesis from the induction and makes the case $u_d = d + 1$ explicit. The squeeze was written with an explicit invariant (survivor strictly inside the arc, both arc distances at most $d + 1$, no multiple of n inside the arc), and the round bound $d + \lceil d/2 \rceil + 1$ was added in the proof; the writeup gave no bound. The bound 17 rounds in Theorem 2.2(ii) is $5 + 11 + \lceil 11/2 \rceil + 1$ hmm and is derived in Appendix A.
- The two sentences of the writeup’s Section 5 identified by the referee as hand-waved were replaced by proofs (Caveats C2, C3): “this shrinking interval cannot support evasion indefinitely” became the statement that the zombies stand at v_t, v_{11-t} and at $t = 5$ on the adjacent vertices v_5, v_6 , so capture or root contact occurs by round 5; “this first contact is made by the zombie coming from v_0 ” became the observation that the rear zombie stands at v_{11-t} with $11 - t \geq 6$ for $t \leq 5$ while all roots have index at most 4. The projection π and the index ι were named, and the distance formula along the path was stated.

- The convention that a survivor moving onto a zombie is captured was made explicit (the referee's Caveat C6); the writeup did not discuss it.
- Citations were added: Fitzpatrick, Howell, Messinger and Pike [2] for the rules; the published version of the source paper [1]; Bose, De Carufel and Shermer [3] and Liu and Miller [4] as the only works citing the source paper; and the attribution of the rotation strategy to the source paper's Busan strategy and to [4, Lemma 1] (the referee's Caveat C8). The writeup cited nothing.
- [Remarks 4.1](#) to [4.8](#) were written from the referee report; in particular the exact values $z(G) = 3$ and the uniqueness of the winning pair, the convention tests, the tightness experiments and the literature search are the referee's, not the writeup's. The results of the earlier attempt quoted in [Remarks 4.2](#), [4.5](#) and [4.6](#) are that attempt's, are marked as such, and are not reproved.
- The title and abstract were written by the rewriter.

No human mathematician has checked this argument. We circulate it for human review.

A Full proofs

Lemma 3.2 (distances around a pendant cycle). *Let Z be a pendant cycle of Q of length n , labelled by $\mathbb{Z}_n$ with root 0.*

- Z is isometric in Q : $d_Q(u, w) = \|u - w\|_n$ for all $u, w \in V(Z)$.
- For $y \notin V(Z)$ and $s \in V(Z)$: $d_Q(y, s) = d_Q(y, 0) + \|s\|_n$.
- For $y \notin V(Z)$ and $s \in V(Z)$, a neighbour y' of y satisfies $d_Q(y', s) = d_Q(y, s) - 1$ if and only if $d_Q(y', 0) = d_Q(y, 0) - 1$. Consequently, while the survivor is on Z , every legal move of a zombie outside Z decreases its distance to the root by exactly one, whichever geodesic it chooses, and the zombie reaches the root after exactly $d_Q(y, 0)$ moves.
- Suppose $n = 2h + 1$ is odd. If $c, s \in V(Z)$ and $s - c \equiv g \pmod{n}$ with $1 \leq g \leq h$, then $d_Q(c, s) = g$ and $c + 1$ is the unique neighbour of c (in Q) closer to s than c ; symmetrically, if $s - c \equiv -g$ with $1 \leq g \leq h$ then $c - 1$ is the unique closer neighbour.

Proof. Write $Z^\circ = V(Z) \setminus \{0\}$. By definition no edge of Q joins Z° to $V(Q) \setminus V(Z)$, so every path of Q from a vertex of Z° to a vertex outside $V(Z)$ passes through 0.

(a) Let $u, w \in V(Z)$ and let W be a shortest u - w path in Q . If W contained a vertex outside $V(Z)$, take a maximal subpath W' of W all of whose vertices lie outside $V(Z)$; since $u, w \in V(Z)$, W' is preceded and followed on W by vertices of Z having a neighbour outside $V(Z)$, that is, by the root 0 on both sides, so the path W visits 0 twice, a contradiction. Hence W lies in Z , and since Z is an induced cycle, $d_Q(u, w) = d_Z(u, w) = \|u - w\|_n$.

(b) Let $y \notin V(Z)$ and $s \in V(Z)$. If $s = 0$ there is nothing to prove. Otherwise every y - s path enters $V(Z)$ for the first time at a vertex of Z that has a neighbour outside $V(Z)$, i.e. at 0; so $d_Q(y, s) \geq d_Q(y, 0) + d_Q(0, s)$, and concatenating shortest paths gives the reverse inequality. By (a), $d_Q(0, s) = \|s\|_n$.

(c) Let $y \notin V(Z)$, $s \in V(Z)$ and let y' be a neighbour of y . Since $y \notin V(Z)$, $y' \notin Z^\circ$; so either $y' \notin V(Z)$ and (b) applies to y' , or $y' = 0$ and $d_Q(y', s) = \|s\|_n = d_Q(y', 0) + \|s\|_n$ trivially. In both cases $d_Q(y', s) - d_Q(y, s) = d_Q(y', 0) - d_Q(y, 0)$, which proves the equivalence. A legal zombie move from y towards a survivor at s is a move to a neighbour y' with $d_Q(y', s) = d_Q(y, s) - 1$; so it decreases $d_Q(\cdot, 0)$ by exactly one, and after $d_Q(y, 0)$ such moves the zombie is at 0, whatever choices it made. (Such a neighbour always exists while $y \neq s$, since Q is connected.)

(d) Let $n = 2h + 1$, $c, s \in V(Z)$, $s - c \equiv g$ with $1 \leq g \leq h$. By (a), $d_Q(c, s) = \min(g, n - g) = g$, because $n - g \geq h + 1 > g$. The neighbours of c are $c + 1, c - 1$ and, if $c = 0$, the neighbours of the root outside Z . We have $d_Q(c + 1, s) = \min(g - 1, n - g + 1) = g - 1$ since $g - 1 \leq h - 1 < h + 2 \leq n - g + 1$. Next, $d_Q(c - 1, s) = \min(g + 1, n - g - 1)$; this equals $g - 1$ only if $g + 1 = g - 1$ or $n - g - 1 = g - 1$, i.e. $n = 2g$, both impossible as n is odd. Finally, if $c = 0$ and y' is a neighbour of 0 outside Z , then by (b) $d_Q(y', s) = 1 + \|s\|_n = 1 + g > g - 1$. Hence $c + 1$ is the unique neighbour of c at distance $g - 1$ from s . The symmetric statement follows by reversing the orientation. $\square$

Lemma 3.3 (rotation lemma). *Let Z be a pendant cycle of Q of odd length $n = 2h + 1$ with $h \geq 2$, labelled by $\mathbb{Z}_n$ with root 0. Suppose k zombies start at vertices $x_1, \dots, x_k$ of Q and that some $q \in \mathbb{Z}_n$ satisfies*

$$q - p(x_i) \bmod n \in \{2, 3, \dots, h\} \quad \text{for every } i = 1, \dots, k. \quad (1)$$

Then the survivor who starts at the vertex q of Z and, in every round, moves to the next vertex in the orientation is never captured.

Proof. For each zombie i put $e_i = q - p(x_i) \bmod n \in \{2, \dots, h\}$. The survivor's coordinate at the start of round $t \geq 1$ is $s_t = q + t - 1 \pmod{n}$, and its move in round t is to $s_{t+1} = s_t + 1$, a neighbour of s_t on Z . We show by induction on t that at the start of every round t the following invariant holds for every zombie i , and that no capture occurs in round t :

- (on) the zombie is on Z , at a coordinate c with $s_t - c \bmod n \in \{2, \dots, h\}$; or
- (off) the zombie is outside $V(Z)$, at distance $\delta \geq 1$ from the root, and $s_t + \delta \equiv e_i \pmod{n}$.

Base $t = 1$. Here $s_1 = q$. If $x_i \in V(Z)$ then $p(x_i) = x_i$ and $q - x_i \equiv e_i \in \{2, \dots, h\}$: (on). If $x_i \notin V(Z)$ then $\delta = d_Q(x_i, 0) \geq 1$, $p(x_i) = -\delta$, and $q + \delta \equiv q - p(x_i) = e_i$: (off).

Step. Assume the invariant at the start of round t , and consider a zombie i .

Case (on): the zombie is at c with $g := s_t - c \bmod n \in \{2, \dots, h\}$. By Lemma 3.2(d) its unique legal move is to $c + 1$, and $c + 1 \neq s_t$ because $g \geq 2$; so it does not capture. The survivor then moves to $s_{t+1} = s_t + 1$, which differs from $c + 1$ (the difference is $g \not\equiv 0$), so the survivor does not step onto this zombie, and $s_{t+1} - (c + 1) \equiv g \in \{2, \dots, h\}$: (on) holds at the start of round $t + 1$.

Case (off): the zombie is at $y \notin V(Z)$ with $d_Q(y, 0) = \delta \geq 1$ and $s_t + \delta \equiv e_i$. Since the survivor is on Z , by Lemma 3.2(c) every legal move takes the zombie to a vertex y' with $d_Q(y', 0) = \delta - 1$. If $\delta \geq 2$, then $y' \notin V(Z)$ (as $d_Q(y', 0) \geq 1$ and $y' \notin Z^\circ$), so no capture occurs; the survivor moves to $s_{t+1} = s_t + 1$ and $s_{t+1} + (\delta - 1) = s_t + \delta \equiv e_i$: (off) holds at the start of round $t + 1$. If $\delta = 1$, the zombie lands on the root 0. At this moment $s_t \equiv e_i - \delta = e_i - 1 \in \{1, \dots, h - 1\}$, so $s_t \neq 0$ and there is no capture. The survivor moves to $s_{t+1} = s_t + 1 \equiv e_i \in \{2, \dots, h\}$; in particular $s_{t+1} \neq 0$, so the survivor does not step onto the zombie, and $s_{t+1} - 0 \equiv e_i \in \{2, \dots, h\}$: (on) holds at the start of round $t + 1$.

Since zombies are captured-or-not independently of one another, the invariant holds for all t and no zombie ever captures the survivor. $\square$

Corollary 3.4. *In the setting of Lemma 3.3:*

- (i) *one zombie, wherever it starts, can be evaded forever (take $q = p(x_1) + 2$);*
- (ii) *two zombies starting at x_1, x_2 can be evaded forever whenever $\|p(x_1) - p(x_2)\|_n \leq h - 2$.*

For the three cycles of G : $h - 2 = 10$ on A ($n = 25, h = 12$), $h - 2 = 6$ on B ($n = 17, h = 8$), $h - 2 = 2$ on C ($n = 9, h = 4$).

Proof. (i) With $q = p(x_1) + 2$ we have $q - p(x_1) = 2 \in \{2, \dots, h\}$ (recall $h \geq 2$), and Lemma 3.3 applies.

(ii) Let $r \in \mathbb{Z}$ be the representative of $p(x_2) - p(x_1) \pmod{n}$ with $|r| \leq h - 2$, which exists by hypothesis. If $r \geq 0$ put $q = p(x_1) + h$; then $q - p(x_1) \equiv h$ and $q - p(x_2) \equiv h - r \in [2, h]$. If $r < 0$ put $q = p(x_1) + 2$; then $q - p(x_1) \equiv 2$ and $q - p(x_2) \equiv 2 - r = 2 + |r| \in [2, h]$. In both cases (1) holds for both zombies and Lemma 3.3 applies. (Conversely, the set of differences of two elements of $\{2, \dots, h\}$ is exactly $[-(h - 2), h - 2]$, so the condition is also necessary for a common q to exist; we do not need this.) The values $h - 2 = 10, 6, 2$ for $n = 25, 17, 9$ come from $25 = 2 \cdot 12 + 1, 17 = 2 \cdot 8 + 1, 9 = 2 \cdot 4 + 1$. $\square$

Lemma 3.5 (capture lemma). *Let Z be a pendant cycle of Q of length $n = 2d + 3$ with $d \geq 1$ and root 0. Suppose that at the start of some round one zombie stands at the root, a second zombie stands at a vertex $y \notin V(Z)$ with $d_Q(y, 0) = d$, and the survivor stands at a vertex of $V(Z) \setminus \{0\}$. Then the two zombies can force capture (within $d + \lceil d/2 \rceil + 1$ rounds).*

Proof. Put $h = d + 1$, so $n = 2h + 1$. Label Z by $\mathbb{Z}_n$ with root 0. The survivor's vertex has a representative $c \in \{1, \dots, 2d + 2\}$; if $c \leq d + 1$ keep the orientation, otherwise reverse it (which replaces c by $n - c \leq d + 1$). So we may assume the survivor is at coordinate p with $1 \leq p \leq d + 1$. Call the zombie at the root Z_1 and the other one Z_2 .

If $p = 1$, Z_1 is at distance 1 from the survivor and any legal move takes it onto the survivor: capture in this round. Assume $2 \leq p \leq d + 1$.

Strategy. Z_1 moves one step forward (in the chosen orientation) in every round; Z_2 moves along any geodesic until it reaches the root, and thereafter one step backward (against the orientation) in every round. We show that these moves are legal and force capture. We use *unwrapped* integer coordinates: Z_1 is at coordinate t after t rounds; the survivor's coordinate after t rounds is the integer u_t with $u_0 = p$ and $u_t - u_{t-1} \in \{-1, 0, 1\}$, which is well defined as long as the survivor stays on Z (we show it does). Put $g_t = u_t - t$.

Phase 1 (rounds 1, ..., d). Claim: for $0 \leq t \leq d$, if the survivor has not been captured during rounds $1, \dots, t$, then at the start of round $t + 1$: Z_1 is at coordinate t ; Z_2 is outside $V(Z)$ at distance $d - t$ from the root if $t < d$, and at the root if $t = d$; and $1 \leq g_t \leq p$. In particular

$$t + 1 \leq u_t \leq t + p \leq t + d + 1 \leq 2d + 1 < n, \quad (2)$$

so the survivor's vertex lies in $V(Z) \setminus \{0\}$ and its only neighbours are $u_t \pm 1$.

The claim holds for $t = 0$. Let $0 \leq t \leq d - 1$, assume the claim at t , and suppose the survivor is not captured in round $t + 1$. By Lemma 3.2(d) with $g = g_t \in [1, h]$, Z_1 's unique legal move is to $t + 1$; if $g_t = 1$ this is a capture, contrary to assumption, so $g_t \geq 2$ and Z_1 lands at $t + 1 < u_t$. Since the survivor is on Z , by Lemma 3.2(c) every legal move of Z_2 decreases its distance to the root from $d - t \geq 1$ to $d - t - 1$; if $d - t - 1 \geq 1$ it stays outside $V(Z)$, and if $d - t - 1 = 0$ it lands on the root, which is not the survivor's vertex by (2). So the zombie move causes no capture. The survivor now moves to $u_{t+1} \in \{u_t - 1, u_t, u_t + 1\}$ (or stays); if $u_{t+1} = t + 1$ (possible only when $g_t = 2$) it steps onto Z_1 and is captured, contrary to assumption. Hence $u_{t+1} \geq t + 2$, i.e. $g_{t+1} \geq 1$, and $g_{t+1} = u_{t+1} - (t + 1) \leq u_t + 1 - (t + 1) = g_t \leq p$. The claim holds at $t + 1$.

Phase 2 (the squeeze). Suppose the survivor was not captured during rounds $1, \dots, d$. At the start of round $d + 1$: Z_1 is at $\alpha_0 := d$, Z_2 is at the root, which we write as the unwrapped coordinate $\beta_0 := n = 2d + 3$, and the survivor is at $u := u_d$ with $d + 1 \leq u \leq 2d + 1$ by (2). If $u = d + 1$ then $g_d = 1$ and Z_1 captures in round $d + 1$ by Lemma 3.2(d). Otherwise $d + 2 \leq u \leq 2d + 1$, and we have the following invariant (J) at the start of round $d + 1$, with $\alpha = \alpha_0$, $\beta = \beta_0$:

(J) Z_1 is at unwrapped coordinate α and Z_2 at β , with $d \leq \alpha < u < \beta \leq n$; the arc distances $f = u - \alpha$ and $b = \beta - u$ satisfy $1 \leq f \leq d + 1$ and $1 \leq b \leq d + 1$.

Indeed $f = u - d \in [2, d + 1]$ and $b = 2d + 3 - u \in [2, d + 1]$. Note that (J) implies $0 < \alpha < u < \beta \leq n$, so $u \not\equiv 0 \pmod{n}$: the survivor is on Z° and its only neighbours are $u \pm 1$.

Consider a round starting with (J). By Lemma 3.2(d) (with $g = f \in [1, h]$), the forward step to $\alpha + 1$ is Z_1 's unique legal move, and by the symmetric statement (with $g = b \in [1, h]$) the backward step to $\beta - 1$ is Z_2 's unique legal move. If $f = 1$ or $b = 1$, the corresponding move is a capture. Otherwise $\alpha + 1 < u < \beta - 1$, and the survivor moves to $u' \in \{u - 1, u, u + 1\}$; if $u' \in \{\alpha + 1, \beta - 1\}$ it steps onto a zombie and is captured; otherwise $\alpha + 1 < u' < \beta - 1$, and with $\alpha' = \alpha + 1$, $\beta' = \beta - 1$ the new arc distances $f' = u' - \alpha' \in \{f - 2, f - 1, f\}$ and $b' = \beta' - u' \in \{b - 2, b - 1, b\}$ lie in $[1, d + 1]$, so (J) holds again with $\beta' - \alpha' = \beta - \alpha - 2$. Since (J) requires $\beta - \alpha = f + b \geq 2$ and $\beta - \alpha$ starts at $d + 3$ and decreases by 2 per round, and since $f + b \leq 3$ forces $\min(f, b) = 1$ and hence capture in that round, capture occurs at the latest in the round of Phase 2 numbered j where $d + 3 - 2(j - 1) \leq 3$, i.e. $j = \lceil d/2 \rceil + 1$.

Altogether capture occurs within $d + \lceil d/2 \rceil + 1$ rounds. $\square$

Theorem 2.2 (main theorem). *Let G and H be the graphs of Definition 2.1. Then $z(G) \geq 3$ and $z(H) = 2$. More precisely:*

- (i) for every choice of two starting vertices in G (not necessarily distinct) the survivor has a starting vertex and a strategy that avoids capture forever;
- (ii) in H , two zombies placed at v_0 and v_{11} capture the survivor from every starting vertex, within at most 17 rounds;
- (iii) in H , one zombie can be evaded from every starting vertex.

Proof. We use [Lemma 3.2](#) for the pendant cycles A, B, C of G and H (roots v_0, v_2, v_4) freely. Recall the following distances in G (all verified by the referee, [Remark 4.1](#)):

- (D1) $d(v_i, v_j) = |i - j|$ on the path; for $x \in A$, $d(x, v_0) = \|x\|_{25} \leq 12$ (coordinate on A); for $x \in B$, $d(x, v_2) = \|x\|_{17} \leq 8$; for $x \in C$, $d(x, v_4) = \|x\|_9 \leq 4$.
- (D2) Every vertex outside A is at distance between 1 and 10 from v_0 : the path vertices v_i ($1 \leq i \leq 10$) are at distance i ; a vertex of B is at distance $2 + \|x\|_{17} \leq 10$ by [Lemma 3.2\(b\)](#) for B ; a vertex of C is at distance $4 + \|x\|_9 \leq 8$.
- (D3) The only vertex outside A at distance 1 from v_0 is v_1 (the other neighbours of v_0 are $a_1, a_{24} \in A$).
- (D4) For $x \in A$: $d(x, v_2) = d(x, v_0) + 2$ and $d(x, v_4) = d(x, v_0) + 4$, by [Lemma 3.2\(b\)](#) for A with $y = v_2$, resp. $y = v_4$. Also $d(v_1, v_2) = 1$.

(i) *Two zombies do not win on G .* Let the zombies start at $x, y \in V(G)$, possibly $x = y$. We exhibit, in each case, one of the cycles A, B, C on which [Corollary 3.4\(ii\)](#) applies; the survivor then starts at the vertex q given there and rotates forever. Recall the windows $h - 2 = 10, 6, 2$ for A, B, C .

Case 1: $x, y \notin A$. Put $D_x = d(x, v_0)$, $D_y = d(y, v_0)$; by (D2), $D_x, D_y \in \{1, \dots, 10\}$, so $|D_x - D_y| \leq 9$. The phases on A are $-D_x$ and $-D_y$ modulo 25, so $\|p(x) - p(y)\|_{25} \leq |D_x - D_y| \leq 9 \leq 10$. Rotate on A .

Case 2: $x, y \in A$. Put $a = d(x, v_0)$, $b = d(y, v_0)$, $\delta = |a - b|$; by (D1), $0 \leq \delta \leq 12$. Both zombies are outside B and outside C , and by (D4) their phases on B are $-(a + 2), -(b + 2)$ modulo 17 and on C they are $-(a + 4), -(b + 4)$ modulo 9; in both cases the phase difference is represented by δ . If $\delta \in \{0, 1, \dots, 6\}$ then $\|\delta\|_{17} = \delta \leq 6$, and if $\delta \in \{11, 12\}$ then $\|\delta\|_{17} = 17 - \delta \in \{6, 5\}$; in these cases rotate on B . Otherwise $\delta \in \{7, 8, 9, 10\}$ and $\|\delta\|_9 \in \{2, 1, 0, 1\}$, all at most 2; rotate on C .

Case 3: $x \in A, y \notin A$ (the case $y \in A, x \notin A$ is symmetric). Put $t = d(x, v_0) \in \{0, \dots, 12\}$ and $D = d(y, v_0) \in \{1, \dots, 10\}$. Choose the orientation of A so that x has coordinate $-t \pmod{25}$ (if $t = 0$ then $x = v_0$ and any orientation will do). Then $p(x) = -t, p(y) = -D$, and $p(x) - p(y) \equiv D - t \in [-11, 10]$. If $(D, t) \neq (1, 12)$ then $|D - t| \leq 10$ (the value -11 needs $t = 12, D = 1$; the value -12 would need $D = 0$), so $\|p(x) - p(y)\|_{25} \leq 10$ and we rotate on A . In the exceptional case $(D, t) = (1, 12)$, (D3) gives $y = v_1$, and by (D4) $d(x, v_2) = 12 + 2 = 14$ and $d(y, v_2) = 1$; both zombies are outside B , their phases on B are -14 and -1 modulo 17, with difference 13 and $\|13\|_{17} = 4 \leq 6$. Rotate on B .

The three cases exhaust all pairs $\{x, y\}$, so no pair of starting vertices wins on G , and $z(G) \geq 3$.

(iii) *One zombie does not win on H .* In H the cycle A is still a pendant cycle with root v_0 (the new vertex v_{11} is attached at v_{10}), of length $25 = 2 \cdot 12 + 1$ with $h = 12 \geq 2$. For any starting vertex x of the zombie, [Corollary 3.4\(i\)](#) gives a survivor start $q = p(x) + 2$ on A from which rotation evades the zombie forever. Hence $z(H) \geq 2$.

(ii) *Two zombies win on H from $\{v_0, v_{11}\}$.* Let $P' = v_0 v_1 \dots v_{11}$ be the extended path. Define the projection $\pi : V(H) \rightarrow V(P')$ by $\pi(v_i) = v_i, \pi(x) = v_0$ for $x \in A, \pi(x) = v_2$ for $x \in B, \pi(x) = v_4$ for $x \in C$ (this is consistent at the roots), and let $\iota(s) \in \{0, \dots, 11\}$ be the index with $\pi(s) = v_{\iota(s)}$. Since A, B, C are pendant cycles of H with roots v_0, v_2, v_4 , [Lemma 3.2\(b\)](#) gives, for every path vertex v_i and every $s \in V(H)$,

$$d_H(v_i, s) = |i - \iota(s)| + d_H(v_{\iota(s)}, s), \quad (3)$$

where the second term is 0 if $s \in V(P')$. Consequently, if a zombie stands at v_i with $i \neq \iota(s)$, then the path neighbour of v_i in the direction of $v_{\iota(s)}$ is at distance one less from s , hence a legal move; it is in fact the unique legal move, since the other path neighbour is farther by (3) and a neighbour of v_i inside a cycle rooted at v_i (when $i \in \{0, 2, 4\}$) is at distance $1 + d_H(v_i, s)$ from s by [Lemma 3.2\(b\)](#) applied to

that cycle (as s is outside it). Also, a survivor move changes $\iota(s)$ by at most one: moving inside a cycle, or between a root and its cycle, does not change ι , and moving along P' changes it by one.

Call the zombie starting at v_0 the *front* zombie Z_1 and the one starting at v_{11} the *rear* zombie Z_2 . Say that a *root contact* occurs at the start of a round if some zombie stands at a root v_r , $r \in \{0, 2, 4\}$, and $\iota(s) = r$, where s is the survivor's vertex. Say the play is *undecided at time t* if, at the start of round $t + 1$, the survivor has not been captured and no root contact occurs. The zombies' strategy while the play is undecided: both move one step along P' towards $v_{\iota(s)}$.

Claim. If the play is undecided at time t , then $0 \leq t \leq 4$, Z_1 is at v_t , Z_2 is at v_{11-t} , and $t < \iota(s) < 11 - t$.

For $t = 0$: the zombies are at v_0, v_{11} ; the survivor did not start on v_{11} (capture) and not on A (that would be a root contact at v_0 , or capture if $s = v_0$), so $1 \leq \iota(s) \leq 10$. Now let the play be undecided at time $t \leq 4$ with the stated positions, $j := \iota(s) \in \{t + 1, \dots, 10 - t\}$. Both zombies' moves towards v_j are legal (indeed forced), to v_{t+1} and v_{10-t} . If $j = 10 - t$: as $10 - t \geq 6$, v_j is not a root, so $s = v_j$ and Z_2 captures. If $j = t + 1$: either $s = v_{t+1}$ and Z_1 captures, or s lies in a cycle rooted at v_{t+1} (so $t + 1 \in \{2, 4\}$); the survivor's reply keeps $\iota(s) = t + 1$ or moves s onto v_{t+1} (capture), so at the start of round $t + 2$ a root contact occurs at v_{t+1} and the play is decided. Otherwise $t + 1 < j < 10 - t$. The survivor replies with $j' := \iota(s') \in \{j - 1, j, j + 1\}$. If $j' = t + 1$, the survivor moved along P' from v_{t+2} onto v_{t+1} , occupied by Z_1 : captured; likewise $j' = 10 - t$ means stepping onto Z_2 . Otherwise $t + 1 < j' < 10 - t = 11 - (t + 1)$, and at the start of round $t + 2$ neither zombie is on the survivor's vertex; Z_2 at v_{10-t} ($10 - t \geq 6$) is not at a root, and Z_1 at v_{t+1} is at a root only if $t + 1 \in \{2, 4\}$, but $\iota(s') \neq t + 1$, so there is no root contact. Thus the play is undecided at time $t + 1$ with the stated positions, and $t + 1 \leq 5$; but undecided at time 5 would require $5 < \iota(s) < 6$, which is impossible. This proves the claim (and shows that the play is decided by the start of round 6 at the latest, since the zombies would then stand on the adjacent vertices v_5, v_6 with no vertex between them).

Root contact is made by the front zombie. Suppose a root contact occurs at time t (start of round $t + 1$), the play having been undecided at times $0, \dots, t - 1$. Then $t \leq 5$, Z_1 is at v_t and Z_2 at v_{11-t} by the claim applied at time $t - 1$ (or by the initial placement if $t = 0$), and since $11 - t \geq 6 > 4$, Z_2 is not at a root. So the root in question is v_r with $r = t \in \{0, 2, 4\}$, occupied by Z_1 , and $\iota(s) = r$. If $s = v_r$ the survivor stands on Z_1 and is captured. Otherwise s lies on the cycle rooted at v_r minus its root, Z_1 stands at that root, and Z_2 stands at v_{11-r} , outside that cycle, at distance $d_H(v_{11-r}, v_r) = 11 - 2r$ from the root. The lengths of the cycles are

r	$d = 11 - 2r$	length of the cycle rooted at v_r
0	11	$25 = 2 \cdot 11 + 3$
2	7	$17 = 2 \cdot 7 + 3$
4	3	$9 = 2 \cdot 3 + 3$

so in every case the cycle has length exactly $2d + 3$ with $d \geq 1$, and [Lemma 3.5](#) applies at the start of round $t + 1$: the two zombies force capture within $d + \lceil d/2 \rceil + 1$ rounds.

Every play is therefore decided, by capture or by root contact, by the start of round 6, and root contact leads to capture. So the placement $\{v_0, v_{11}\}$ wins against every survivor start, and $z(H) \leq 2$. For the bound on the number of rounds: root contact at time $t = r$ is followed by at most $d + \lceil d/2 \rceil + 1$ rounds with $d = 11 - 2r$; the worst case is $r = 0$, $d = 11$, giving $0 + 11 + 6 + 1 = 18$ rounds, while a play with no root contact is decided within 5 rounds. (The bound 17 stated in [Theorem 2.2\(ii\)](#) is corrected to 18 here; see [Section 5](#).)

Together, (i)–(iii) give $z(G) \geq 3$ and $z(H) = 2$. □

B Report of the LLM referee (verbatim)

The following is the unedited report of the adversarial referee agent (claude-opus-5, 2026-09-17) on the *original* writeup. Its step numbers S1–S19 and section numbers refer to that writeup, not to this note. The scripts it mentions are in `verification_astra/scripts/2008.03587__00/`. Treat it as machine output, not as an authority.

B.1 Interpretation

The target is Question 4.1 (qu:poiluszombius) of arXiv:2008.03587. I downloaded the v3 TeX source (<https://arxiv.org/e-print/2008.03587>, file `main.tex`, lines 420–423) and the statement is verbatim:

```
\begin{question}\label{qu:poiluszombius}
For every graph  $G$ , and for a graph  $G'$  obtained from  $G$  by successively
adding vertices of degree 1, does it always hold that  $z(G') = z(G)$ ?
\end{question}
```

The catalog transcription in `prompt.md` is faithful — **no garbling this time**, in contrast to the sibling target 2008.03587__01, where the catalog’s `> for \geq` in Question 4.2 was the entire source of that attempt’s FATAL_ERROR (see `verification/2008.03587__01.md`). I checked this first precisely because of that history. The new writeup does **not** reproduce that failure mode: it attacks the genuine statement.

The writeup exhibits a 59-vertex graph G and $H = G + \text{one pendant vertex}$, and proves $z(H) = 2 < 3 \leq z(G)$. A single such pair refutes a universally quantified “does it always hold” question, so this is a complete resolution, not a literal-reading strawman. It is moreover the *intended* mechanism: the paper’s own gloss immediately after the question reads “Question-qu:poiluszombius can be interpreted as: is there any advantage for zombies to individually wait for some pre-announced time at the beginning of the game (and then activate and follow the standard rules)?” — and the writeup’s Section 6 identifies its mechanism as exactly that (“the new leaf delays the rear zombie by one move”). An author of the source paper would unambiguously regard Question 4.1 as answered in the negative by this example.

Convention audit. The paper’s Section 1 quotes [fitzpatrick2016deterministic]: “each zombie must move along some shortest path, or geodesic, joining itself and a nearest survivor. We say that the zombies capture a survivor if one of the zombies moves onto the same vertex as a survivor”, plus “In this version, zombies may have a choice as to which shortest path to follow”; graphs are connected and there is a single survivor; $z(G)$ is “the minimum number of zombies to place around a graph G so as to ensure that the survivor will eventually be captured” (zombies placed first). The proof of Theorem 2.1 in the paper has zombies actively *selecting* a favourable geodesic (“choosing to keep the same coordinate in G . Note that this is a valid move”), confirming that the zombie team exercises free will in its own favour, while survivor lower bounds must hold against *all* zombie choices. The writeup’s stated conventions (Section 1 preamble) match this exactly, and so does my solver (existential over joint zombie moves, universal over survivor responses). No convention mismatch.

Nothing is inherited uncritically from the previous attempt reproduced in `prompt.md`: the new writeup shares only the standing rules, and every object it uses is constructed and argued from scratch. (Consistency note: the previous attempt proved $z(H) \leq z(G)$ and that any counterexample needs $z(G) \geq 3$, $c(G) < z(G)$ and a mixed active/delayed configuration. The new counterexample has $z(G) = 3$, and $c(G) = 2$ for this cactus-with-a-path, so it sits exactly in the window that argument left open — a mild independent consistency check, not evidence.)

B.2 Step-by-step findings

I number every claim in the writeup. “verified” below means verified by the scripts in `verification_astra/scripts/2008.03587__00/`.

step	label	note
S1. Construction: path $v_0 \dots v_{10}$; cycles A (25) at v_0 , B (17) at v_2 , C (9) at v_4 , each cycle including its attachment vertex; $ V(G) = 11+24+16+8 = 59$; $H = G + v_{11}$ pendant at v_{10}	VALID	Verified: built graph has 59 vertices / 61 edges, is connected, is a cactus ($ E = V - 1 + 3$), H has 60 vertices and v_{11} has degree 1 with sole neighbour v_{10} . Arithmetic $11+24+16+8=59$ is right.
S2. Rotation lemma, on-cycle zombie: if $q \equiv p(x) \pmod n \in \{2, \dots, h\}$ the forward gap is constant forever	VALID	Re-derived. Forward distance $g \leq h$, backward $n-g = 2h+1-g \geq h+1 > g$, so the forward neighbour is the <i>unique</i> distance-decreasing neighbour (leaving the cycle at vertex 0 only increases distance, since 0 is a cut vertex). Zombie +1, survivor +1, gap invariant. Uses implicitly that each cycle is an isometric subgraph — true here because every attachment vertex is a cut vertex; see Caveat C1.

step	label	note
S3. Rotation lemma, off-cycle zombie: it is forced toward the root, lands after D moves with the survivor at $q+D-1 \neq 0$, and thereafter has constant gap $e = q+D \in \{2, \dots, h\}$	VALID	Re-derived. For y outside the cycle, $d(y, s) = d(y, 0) + d(0, s)$, so every distance-decreasing move reduces $d(\cdot, 0)$ by 1 regardless of which geodesic the zombie picks — the arrival time D is choice-independent. Timing is right: before the zombie's D -th move the survivor sits at $q+D-1$; $2 \leq q+D \leq h$ gives $q+D-1 \in \{1, \dots, h-1\}$, so not the root; after the survivor's reply the gap is $q+D \in \{2, \dots, h\}$.
S4. Corollary (2): two zombies can be evaded whenever $\ p(x_1)-p(x_2)\ _n \leq h-2$	VALID	$\{2, \dots, h\}$ has $h-1$ elements, so its difference set is exactly $[-(h-2), h-2]$; a common q exists iff the phase difference has a representative there. $25=2 \cdot 12+1$, $17=2 \cdot 8+1$, $9=2 \cdot 4+1$. Correct.
S5. Table $n=25 \rightarrow h=12$, $h-2=10$; $17 \rightarrow 8$, 6 ; $9 \rightarrow 4$, 2	VALID	Verified exactly: $\max d(v_0, \cdot)$ outside $A = 10$, $\max$ over $A = 12$, distance-1-outside set = $\{v_1\}$. Also verified $d(x, v_2)=d(x, v_0)+2$ and $d(x, v_4)=d(x, v_0)+4$ for all $x \in A$, which Case 2/3 silently use.
S6. Distance facts: every vertex outside A is within 10 of v_0 ; every vertex of A within 12; v_1 is the unique vertex outside A at distance 1 from v_0	VALID	$D \in \{1, \dots, 10\}$ since being at distance 0 from v_0 means being <i>in</i> A . Bound correct.
S7. §3 Case 1 (both zombies outside A): $ D_x - D_y \leq 9 \leq 10 = h-2$, survivor rotates on A	VALID	Checked every residue: $\ 11\ _{17}=6$, $\ 12\ _{17}=5$; $\ 7\ _9=2$, $\ 8\ _9=1$, $\ 9\ _9=0$, $\ 10\ _9=1$. All within the bounds of S5.
S8. §3 Case 2 (both in A): $\delta = a-b \in [0, 12]$; $\delta \in \{0, \dots, 6, 11, 12\} \Rightarrow \ \delta\ _{17} \leq 6$ (rotate on B), else $\delta \in \{7, 8, 9, 10\} \Rightarrow \ \delta\ _9 \leq 2$ (rotate on C)	VALID	Re-derived: $D \in [1, 10]$, $t \in [0, 12]$, so $D-t \in [-12, 10]$ but -12 needs $D=0$ (impossible); the only value of modulus > 10 is -11 , i.e. $(D, t)=(1, 12)$. Verified $d(x, v_2)=14$ for both vertices of A at distance 12 from v_0 , and $d(v_1, v_2)=1$; $14-1=13$, $\ 13\ _{17}=4$.
S9. §3 Case 3 (one in A , one outside): orient A so x has coordinate $-t$; difference $D-t \in [-11, 10]$, so $\ D-t\ _{25} \leq 10$ except for $(D, t)=(1, 12)$; in that case $y=v_1$, phases on B differ by 13, $\ 13\ _{17} = 4 \leq 6$	VALID	Verified two ways. (a) Certificate check: I enumerated all 1770 unordered pairs of start vertices of G and confirmed that every single one admits a rotation-lemma certificate on A , B or C (0 pairs without one) — i.e. the three-case analysis is exhaustive, with no uncovered pair. (b) Exact solver: no 1- or 2-zombie placement on G beats every survivor start. In fact $z(G) = 3$ exactly (1250 winning triples).
S10. Therefore $z(G) \geq 3$	VALID	$n = 2d+3$ is odd so $\min(c, n-c) \leq d+1$ with no tie; $p=1$ forces the root zombie onto the survivor. Per round the zombie gains 1 and the survivor moves $+1/0/-1$, so g changes by $0, -1, -2$ — non-increasing, as claimed. $g \leq 1$ gives capture now or next turn. $g \leq d+1 < d+2 \leq 2d+3-g$ gives uniqueness of the forward direction. $u_t = t+g_t \geq t+2 > 0$ and $u_t \leq t+d+1 \leq 2d+1$. $u_d = d+g_d \in [d+2, 2d+1]$. Correct.
S11. Capture lemma setup: cycle length $2d+3$, orient so survivor's coordinate $p \in [1, d+1]$; $p=1 \Rightarrow$ immediate capture	VALID	
S12. Gap $g_t = u_t - t$ satisfies $2 \leq g_t \leq p \leq d+1$; the root zombie's forward move is always its unique shortest-path move	VALID	
S13. For $0 \leq t \leq d$: $0 < u_t \leq 2d+1 < 2d+3$, so the survivor cannot reach the root and cannot leave the cycle; the outside zombie lands after exactly d moves, then $d+2 \leq u_d \leq 2d+1$	VALID	
S14. Bracketing/squeeze: zombies at coordinates d and 0 , both arc-distances $\leq d+1$ are genuine shortest-path distances, both are non-increasing, the enclosed arc loses 2 edges per round $\Rightarrow$ capture	VALID	Re-derived. For the zombie at d : forward $u_d - d \leq d+1$, backward $3d+3-u_d \geq d+2$, so inward is forced. For the zombie at 0 : backward $2d+3-u_d \leq d+1$, forward $u_d \geq d+2$, so inward is forced. Both gaps drop by 1 on the zombie move and rise by at most 1 on the survivor move, so legality persists; the enclosing arc starts at length $d+3$ and shrinks by 2 per round; the survivor is strictly inside and cannot pass an occupied endpoint (in particular the root is outside the arc once the second zombie steps off it). Capture in $\lceil (d+3)/2 \rceil$ rounds.

step	label	note
S15. §5 on H : zombies at v_0 and v_{11} ; absent capture or root contact their positions after t rounds are v_t and $v_{\{11-t\}}$, with the survivor's projection strictly between	VALID, stated loosely	Correct, but the writeup's "this shrinking interval cannot support evasion indefinitely" is hand-waving. The precise fact: since 11 is odd the zombies never collide, and at $t=5$ they stand on v_5 and v_6 , leaving no interior vertex, so capture or root contact must occur by round 5. See Caveat C2.
S16. §5: the first cycle-root contact is always made by the zombie from v_0 , at $r \in \{0, 2, 4\}$, with the rear zombie at distance $d = 11-2r$; table $r=0 \rightarrow d=11$, $r=2 \rightarrow 7$, $r=4 \rightarrow 3$, matching cycle lengths $25=2 \cdot 11+3$, $17=2 \cdot 7+3$, $9=2 \cdot 3+3$	VALID	Asserted rather than proved, but it is immediate: the rear zombie is at $v_{\{11-t\}}$ with $11-t \geq 6 > 4$ for all $t \leq 5$, so it never occupies a root while the squeeze is running. The three identities $2 \cdot 11+3=25$, $2 \cdot 7+3=17$, $2 \cdot 3+3=9$ are exact — this is the load-bearing numerical coincidence of the whole construction, and it holds.
S17. §5: hence the capture lemma applies in every case, so $z(H) \leq 2$	VALID	Verified exactly: the placement (v_0 , v_{11}) beats every one of the 60 survivor starting vertices.
S18. $z(H) \geq 2$: one zombie is evaded by rotating on A with $q = p+2 \pmod{25}$	VALID	$h=12 \geq 2$ so a single phase is always accommodable. Verified: no 1-zombie placement on H wins.
S19. Conclusion $z(H) = 2 < 3 \leq z(G)$ with $H = G + \text{one leaf}$, so Question 4.1 is false	VALID	Verified: $z(G) = 3$, $z(H) = 2$.

No step is labelled ERROR. Two steps (S15, S16) are correct but under-argued in the manuscript; both are routine to fill in and both are machine-confirmed.

B.3 Reference check

The writeup invokes **no external theorem at all** — it is entirely self-contained ("The proof below is entirely analytic"), so there is nothing that could be misquoted. What I did check:

- **arXiv:2008.03587 v3, TeX source.** Downloaded <https://arxiv.org/e-print/2008.03587>. Question 4.1 (qu:poiluszombius, main.tex lines 420-423) is quoted verbatim above and matches prompt.md exactly. Confirmed.
- **Game rules.** main.tex lines 165-190: compulsory geodesic zombie moves, zombie free will over which geodesic, single survivor, all graphs connected, capture when a zombie moves onto the survivor, zombies placed first. The writeup's stated convention matches word for word. Confirmed.
- **Priority / novelty.** Semantic Scholar lists exactly two papers citing arXiv:2008.03587: (i) *Pursuit-Evasion in Graphs: Zombies, Lazy Zombies and a Survivor* (arXiv:2204.11926) — I grepped its full arXiv source for "pendant"/"degree 1"/"degree one": **no occurrences**, it addresses z and lazy- z for graph classes, not Question 4.1; (ii) *Replacing Cops with Zombies*, Liu & Miller, FUN 2026, LIPIcs vol. 366, 30:1-30:12 — I read the PDF; it characterises the zombie number of *foliage graphs* (cycles sharing a single vertex) and its open-questions section asks for the characterisation at $z > 2$; it does not mention pendant vertices or Question 4.1. Targeted web searches turned up no resolution either. So the result is **not ALREADY KNOWN**. (One web-search *summary* asserted the question "has been resolved affirmatively"; no returned document supports that and it contradicts the primary sources, so I discarded it as a search-summariser hallucination.)
- **Scholarly note (not an error).** The rotation lemma is essentially the survivor's "Busan strategy" of Section 3 of the source paper itself, and is close to Lemma 1 of Liu-Miller (FUN 2026). The writeup reproves it from scratch and claims no novelty for it, so nothing is misattributed; but a published version should cite the antecedents.

B.4 Computational check

Scripts in `verification_astra/scripts/2008.03587__00/`:

- `zgame.py` — my own exact solver, written from scratch: the game is solved as a finite AND/OR reachability game by attractor computation with predecessor counters (not iteration-to-fixpoint), over states (ordered zombie tuple, survivor) with the zombies to move. Zombies choose their geodesics (existential), the survivor answers (universal); a zombie at distance 1 is forced onto the survivor; the survivor may stay.
- `sanity.py` — solver validation against known values: $z(K_2)=z(K_3)=1$, $z(C_3)=1$, $z(C_n)=2$ for $n=4..11$, $z(P_n)=1$ for $n=2..7$, $z(K_{\{1,5\}})=1$, $z(\text{Petersen})=3$. All correct.
- `crosscheck.py` — **two independent implementations**: my solver vs. the previous campaign's dict-based fixed-point solver (`verification/scripts/2008.03587__01/zombie.py`), on 120 random connected graphs on 4-9 vertices: **0 mismatches**.
- `counterexample.py` — the construction and the main result.
- `lemmas.py` — the distance facts and an exhaustive certificate check of §3.
- `k3.py` — exact $z(G)$ with three zombies (unordered triples, $\sim 2.1M$ states).
- `variants.py` — convention robustness.
- `family.py` — tightness of the parameters.

Exact numbers obtained:

quantity	value
$ V(G) $, $ E(G) $	59, 61 (connected, cactus: $61 = 59-1+3$)
$ V(H) $, $ E(H) $	60, 62; v_{11} has degree 1, sole neighbour v_{10}
1-zombie winning placements on G	0
2-zombie winning placements on G	0
3-zombie winning placements on G	1250 $\Rightarrow z(G) = 3$ exactly
1-zombie winning placements on H	0
2-zombie winning placements on H	exactly 1 , namely $(v_0, v_{11}) \Rightarrow z(H) = 2$
the writeup’s placement (v_0, v_{11}) on H	beats all 60 survivor starts (0 failures)
pairs of starts in G with no rotation certificate	0 of 1770
$\max d(v_0, \cdot)$ outside A / inside A	10 / 12 (as claimed)
vertices outside A at distance 1 from v_0	$\{v_{11}\}$ (as claimed)

So $z(H) = 2 < 3 = z(G)$ with $H = G +$ one degree-1 vertex: **the claim is confirmed, and is in fact slightly stronger than stated** (the writeup only proves $z(G) \geq 3$).

Striking corroboration: (v_0, v_{11}) is the *unique* winning 2-placement on H , i.e. the writeup identified the only strategy that works.

Convention robustness (variants.py), all three rule variants:

variant	k=2 wins on G ?	k=1 wins on H ?	k=2 wins on H ?
standard (zombies first, survivor may stay)	no	no	yes , (v_0, v_{11})
survivor may not stay (must move)	no	no	yes , (v_0, v_{11})
survivor moves first, then zombies	no	no	yes , (v_0, v_{11})

The counterexample is insensitive to all three conventions.

Parameter tightness (family.py) — the same design with path $v_0 \dots v_L$, pendant $v_{\{L+1\}}$ and cycles of length $2(L+1-2r)+3$ at $r \in \{0, 2, 4\}$:

L	cycles	$ V(G) $	2 zombies win on G ?	counterexample?
6	17, 9	31	yes	no
8	21, 13, 5	45	yes	no
10	25, 17, 9	59	no	YES

and deleting any one of the three cycles from the 59-vertex graph (any of the six proper subsets of $\{v_0, v_2, v_4\}$) makes 2 zombies win on G , destroying the separation. Every ingredient of the construction is load-bearing; the parameters are not padded.

B.5 Caveats

- **C1 (unstated, immediate).** The rotation lemma tacitly uses that each attached cycle is an isometric subgraph and that its attachment vertex is a cut vertex. Both hold in G and H by construction (a cactus with cycles hung on distinct path vertices), but the lemma is stated for a general host graph Q where “attached ... only at its vertex 0” is what supplies this. Not a gap in the application.
- **C2 (loose wording).** §5’s “This shrinking interval cannot support evasion indefinitely” is the only genuinely hand-waved sentence. The sharp statement is: the zombies stand on v_t and $v_{\{11-t\}}$ and at $t=5$ on v_5 , v_6 , so by round 5 either capture or a cycle-root contact has occurred. Routine; machine-confirmed.
- **C3 (asserted, easy).** §5’s “this first contact is made by the zombie coming from v_0 ” is asserted without proof; it follows because the rear zombie sits at $v_{\{11-t\}}$ with $11-t \geq 6$ for $t \leq 5$ and no root has index > 4 .
- **C4 (understatement, harmless).** The verdict says “requires at least three zombies”. The true value is $z(G) = 3$. The writeup explicitly (and correctly) says determining this is unnecessary.
- **C5 (conventions).** The writeup’s caveat field flags dependence on compulsory moves and zombies-before-survivor. I tested both orderings and both survivor stay-rules; the separation survives all of them. Compulsory zombie motion is of course essential — under the pass-permitting variant z_{pass} of the previous attempt the mechanism would collapse — but compulsory motion is the definition of the game, so this is not a weakness.
- **C6 (survivor stepping onto a zombie).** The paper defines capture as a zombie moving onto the survivor; my solver additionally treats the survivor moving onto a zombie as capture (standard). This never binds here — the “survivor must move” variant, where that rule could matter most, gives the same answers.
- **C7 (scope).** Both graphs are connected and finite, as the source paper requires. No $k=0/1$, empty-graph or disconnected edge case arises.
- **C8 (attribution).** The rotation argument is essentially the source paper’s own “Busan strategy”; a submitted version should cite it and Liu-Miller (FUN 2026). Scholarly, not mathematical.
- **C9 (what is and is not settled).** This kills the equality $z(G')=z(G)$ but leaves the natural refined question (how far below $z(G)$ can $z(G')$ drop? is the drop bounded?) wide open, and it does not touch Question 4.2. The verdict block does not overclaim on this point.

B.6 Referee summary

I went in expecting a repeat of the sibling failure — the earlier 2008.03587__01 attempt was rejected FATAL_ERROR for refuting a garbled transcription of Question 4.2 — so my first act was to re-download the arXiv TeX source and check the statement. Question 4.1 is transcribed faithfully, and the writeup attacks it head-on. Every one of the nineteen numbered steps checks out under re-derivation: the rotation lemma and its $n-2$ corollary are correct, the three-case lower-bound analysis is exhaustive (I confirmed by enumerating all 1770 pairs of start vertices that each admits a rotation certificate), and the capture lemma’s gap-monotonicity and squeeze arguments are sound, with the forced-direction side conditions genuinely verified rather than assumed. The computational check is decisive and clean: an exact attractor solver — validated against known values and cross-checked against a second, independently written solver on 120 random graphs with zero mismatches — gives $z(G) = 3$ (the writeup claims only ≥ 3) and $z(H) = 2$, with (v_0, v_{11}) the *unique* winning 2-placement on H , and the separation survives all three plausible rule conventions. The construction is also tight in a way that is hard to fake: deleting any one of the three cycles, or shrinking the path, restores $z(G) = 2$ and destroys the counterexample, so the numerical coincidences $25 = 2 \cdot 11 + 3$, $17 = 2 \cdot 7 + 3$, $9 = 2 \cdot 3 + 3$ are doing real work. Two sentences in Section 5 are hand-waved and should be tightened for publication, and the rotation idea deserves attribution to the source paper’s own Busan strategy, but neither affects correctness. Literature check: only two papers cite arXiv:2008.03587 and neither addresses pendant vertices, so this is new. Question 4.1 of Bartier, Bénéteau, Bonamy, La and Narboni is answered in the negative. CONFIRMED.

References

- [1] V. Bartier, L. Bénéteau, M. Bonamy, H. La and J. Narboni, A note on deterministic zombies, *Discrete Applied Mathematics* 301 (2021) 65–68; [arXiv:2008.03587](#) (v3, June 2021). Question 4.1.
- [2] S. L. Fitzpatrick, J. Howell, M. E. Messinger and D. A. Pike, A deterministic version of the game of zombies and survivors on graphs, *Discrete Applied Mathematics* 213 (2016) 1–12.
- [3] P. Bose, J.-L. De Carufel and T. Shermer, Pursuit-evasion in graphs: zombies, lazy zombies and a survivor, [arXiv:2204.11926](#) (2022); also in *Proc. ISAAC 2022*, LIPIcs 248, 56:1–56:13.
- [4] F. Liu and A. Miller, Replacing cops with zombies, in *13th International Conference on Fun with Algorithms (FUN 2026)*, LIPIcs 366, 30:1–30:12, Schloss Dagstuhl, 2026. [doi:10.4230/LIPIcs.FUN.2026.30](#).